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The Decay Algebra: From Export Ledgers to a Continuum Entropy-Production Density

Paper C (decay algebra) — **Draft Status:** 2026-07-30 · Preliminary research · self-contained, classical, no gravitational claims **Scope contract:** [../DELIVERY_SCOPE.md](#) · **Depends on:** Paper A [../01-foundations/PAPER_A_computational_entropy.md](#) **Canonical notes:** [synthesis/m11f-m11i](#) · **Reproducible:** [simulations/classical/m11_{idem,decay_algebra,decay_algebra}](#)

Abstract

Paper A defined computational entropy as the entropy of a map’s output distribution and showed that the entropy exported by an irreversible computation, $H(X \rightarrow Y)$, is Landauer-exact but **path-dependent** and **non-additive**. That raises a sharp question: does a *coupled* lattice of local maps have a well-defined export **density** at all, and if so can it be computed without exponential enumeration? We answer both. First, the IDEM *decay vector* gives a tight upper bound on export, exact for maps with product (fully-recoverable) preimages. Second, promoting the hard decay flag to a **belief transported across the coupling boundary** yields a *decay algebra* — a transfer operator that computes the exact coupled export density. We prove a density theorem for the 1D nearest-neighbour case, generalize it to arbitrary gates and window widths (with a clean dichotomy: the algebra is exactly finite iff the gate has a fully-recoverable branch), lift it to a 2D strip transfer, and finally show a **hydrodynamic limit**: with a smoothly varying spatial field the discrete density converges (at rate $O(1/N)$) to a **constructive continuum entropy-production density field** ($\mathbf{x}$). Every result is backed by an exactly-reproducing computation. We make **no** claim that is a gravitational load term; that identification is discussed only as an explicit, deferred semantic bridge.

1. Setup: from a path-dependent ledger to a density question

For a computation $X \rightarrow Y$ on uniform inputs, Paper A’s **export identity** is

$$H(X) - H(Y) = H(X \mid Y) =: \text{export},$$

the chain-rule remainder (AND gate: $H(Y)=0.811278$, export $=1.188722$). Paper A also proved two facts that make a continuum density non-obvious: cumulative export ΣL_S is **path-dependent** (Thm 3), and the soft-recoverability slot is **non-additive** (Thm F).

So for a lattice of *coupled* local maps (maps sharing input wires), it is not clear that the total export is even extensive. We fix a concrete family: a sliding window of width w over an i.i.d. fair bit stream,

$$Y_i = g(b_i, \dots, b_{i+w-1}),$$

and study the per-site export density $a := \lim_{k \rightarrow \infty} E(k)/k$, where $E(k) = H(X_{Y_{0:k-1}})$.

Extensivity witness (`m11_lattice_export_density.py`). For coupled AND lattices $E(k)$ is linear in k — a well-defined bulk density exists (shared-input $a 0.30076$, residual $\sim 10^{-5}$; chained $a 0.999$) — and the path-dependence of Paper A saturates to an $O(1)$ **boundary** term, so per-site it vanishes like $1/k$. The obstruction is real but subleading: a density exists.

2. The decay vector bounds export (exact for product maps)

The IDEM *decay vector* records, per input coordinate and per output branch, whether that input is recoverable from Y ($d_i=0$) or lost ($d_i=1$). Since $X(Y=y)$ is uniform on the preimage $f^{-1}(y)$,

$$H(X | Y) = \sum_y p(y) \log_2 |f^{-1}(y)| \leq \sum_y p(y) \#\{\text{unrecoverable coords}\} = \text{decay bound},$$

with **equality iff every preimage is a product subcube** (a fully-recoverable branch). This is proved and asserted over a map family (`m11_idem_export_density.py`):

map	exact export	decay bound	gap	product?
<code>erase(2,1)</code> , <code>erase(3,1)</code> , <code>erase(4,2)</code>	= bound	= exact	0	
<code>AND(2)</code>	1.18872	1.50000	0.311	
<code>majority(3)</code> , <code>parity(3)</code>	2.00000	3.00000	1.000	

Consequence (utility). For product/erasure lattices the export density is read directly from local decay metadata in $O(N)$ — enumeration-free and exact. For non-product/coupled maps the bound is loose (e.g. the shared-input AND lattice has true density $0.30076 \approx 1.5$), which motivates the algebra of §3.

3. The decay algebra and the 1D density theorem

The looseness of the hard bound has two sources — fractional entropy of non-product preimages, and correlations from shared wires. Both are cured by replacing the $\{0,1\}$ flag with a **belief over the shared boundary bit that is transported site-to-site** (a transfer operator).

Theorem 1 (1D density theorem; m11f). *For the 1D nearest-neighbour AND lattice $Y_i = b_i \oplus b_{i+1}$, the density*

$$a = \lim_k E(k)/k \text{ exists and } a = 1 - h_Y, \quad h_Y = \sum_{r \geq 0} \rho(r) h_2(p_1(r)),$$

with $E(k) = ak + b + o(1)$. The belief filter $\beta_i = \Pr(b_i=1 | Y_{0:i})$ resets to 1 on $Y_i=1$, so it collapses to a **run-length Markov chain** with $\beta_0=1$, $\beta_{r+1} = (1 - \beta_r)/(2 - \beta_r)$, $p_1(r) = \beta_r/2$, stationary

law $(r)_{j < r} (1 - p_1(j))$. Fixed point $= (3 - \sqrt{5})/2$, density

$$a = 0.3007568.$$

Proof idea. (Y_i) is a sliding-block factor of i.i.d. stationary ergodic the entropy rate exists (Shannon–McMillan–Breiman); the input rate is exactly 1, giving $a = 1 - h_Y$. The filter is a sufficient statistic, and its reset structure makes the run-length chain ergodic with the stated stationary law. The belief-transfer recursion reproduces the $O(2^k)$ enumeration to $< 10^{-6}$ at $O(R)$ cost (`m11_decay_algebra.py`).

4. General- w theorem: any gate, and the reset dichotomy

Theorem 2 (general w ; m11g). *For any gate $g: \{0, 1\}^w \rightarrow \{0, 1\}$ and the sliding-window lattice:*

1. **(Existence.)** (Y_i) is $(w-1)$ -dependent (outputs w apart use disjoint bits), hence stationary ergodic; $a = 1 - h_Y$ exists with $E(k) = ak + O(1)$.
2. **(Blackwell representation.)** $h_Y = -\sum_{\mathbf{y}} p(\mathbf{y}) \log p(\mathbf{y})$, where p is the unique invariant measure of the belief-transfer operator on the 2^{w-1} -state boundary simplex.
3. **(Regeneration.)** If g is a **reset gate** — some output value has a *singleton* preimage (a fully-recoverable $[0, \dots, 0]$ decay branch) — the filter restarts at the same point mass on every reset, so the inter-reset cycles are i.i.d. and h_Y is given by **renewal-reward**; the algebra is then exactly finite.

Corollary (dichotomy). The belief-transfer algebra is exactly finite iff g is a reset gate. This is executably verified (`m11_decay_algebra_general.py` asserts `exact_singleton-preimage`):

gate	w	reset?	method	density a
AND	2,3,4	yes	exact	0.30076 / 0.56568 / 0.74503
OR	2,3	yes	exact	0.30076 / 0.56568
threshold 2	3	no	Monte Carlo	0.23734

Thus the *decay-vector structure of a single gate* — whether some branch fully recovers its inputs — controls whether the *composition algebra* is finitely exact. The reset direction is proved; the converse (non-reset non-atomic) is left conjectural.

5. 2D strip transfer

Lifting to a 2D lattice (2×2 -plaquette AND, $Y_{ij} = b_{i,j} b_{i,j+1} b_{i+1,j} b_{i+1,j+1}$), the boundary between processed and unprocessed sites becomes a **row-cut of width W** and the transfer becomes an operator on the 2^W cut configurations — the Bayesian-filter form of a statistical-mechanics strip transfer matrix.

Result (m11h). The 2D per-site density exists (finite-dependence on 2^2 ; proved). The width- W density $a_W = (W - h_{\text{row}}) / (W - 1)$ computed by the strip operator matches exact enumeration (`m11_decay_algebra_2d.py`):

W	2^W	a_{enum}	a_{transfer}	$ \Delta $
2	4	1.68412	1.68429	1.7e-4
3	8	1.19251	1.19329	7.8e-4
4	16	—	1.03098	—

a_W converges ($1.684 \rightarrow 1.193 \rightarrow 1.031$). The row-cut carries 2^W states and 2^{W-1} emission branches, so both enumeration and transfer cost grow exponentially in strip width. This exponential wall is generic in $d \geq 2$ and is exactly why a genuine limit — not more enumeration — is required.

6. Continuum embedding: a density field

Let the local statistics vary in space: $b_j \text{Bernoulli}(x_j)$, $x_j = j/N$, for a smooth field $\phi : [0,1] \rightarrow (0,1)$ with the 1D nearest-neighbour AND coupling. The homogeneous density at bias ϕ is $a(\phi) = h_2(\phi) - h_Y(\phi)$.

Result (local equilibrium; m11i). As $N \rightarrow \infty$, the coupled per-site density converges to the *pointwise* homogeneous density,

$$\sigma(x) = a(\theta(x)) = h_2(\theta(x)) - h_Y(\theta(x)) \quad \text{at rate } O(1/N).$$

The belief filter mixes geometrically (mixing length $O(1)$ sites), so its law at site i equals the homogeneous stationary law at (x_i) up to the bias variation over one mixing length, $O(1/N)$. Witness (`m11_continuum_embedding.py`, $(x) = 0.35 + 0.30x$, interior discrepancy $\Delta(N)$):

N	40	80	160	320	640
$\Delta(N)$	2.54e-3	1.72e-3	1.00e-3	5.34e-4	2.75e-4

Halving ratios $1.48 \rightarrow 1.71 \rightarrow 1.88 \rightarrow 1.95 \rightarrow 2$, and $\Delta \cdot N$ approaches a constant, i.e. $\Delta(N) = O(1/N)$. The discrete export ledger therefore admits a **constructive continuum entropy-production density field** (x) , with total $\int_0^1 dx = 0.29937$ for the sample field (profile $\phi = 0.446, 0.379, 0.301, 0.220, 0.146$ at $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$).

7. Scope, rigor, and non-claims

Statement	Rigor
Coupled export is extensive; density exists	witnessed
Decay vector upper-bounds export, exact for product maps	proved
1D density theorem $a = 1 - h_Y = 0.3007568$ (Thm 1)	proved
General- w existence + Blackwell + reset-regeneration (Thm 2)	proved (non-atomicity converse conjectural)
2D density exists; width- W strip transfer	proved (existence); convergence numerical
Continuum density field $(x) = a(\phi(x))$, $O(1/N)$	witnessed (general theorem open)

Explicit non-claims. $(\mathbf{x})$ is a **classical** entropy-production density for a specific lattice family. It is **not** identified with the gravitational load term $|\mathrm{d}\mathbf{S}_c/\mathrm{d}\mathbf{l}|$, **not** continuum $L(\mathbf{g})$, and carries **no** gravitational content. A general hydrodynamic theorem (all gates, 2D) is open. The map from to any physical load term is, at most, a separate labeled semantic step and is not undertaken here.

Takeaway. From the Landauer-exact export ledger, promoting the decay vector to a transported belief yields a compositional *decay algebra* that computes coupled export densities exactly (for reset gates) and, in a hydrodynamic limit, a constructive continuum entropy-production density field. IDEM’s decay vector thereby acquires a proved, load-bearing role: its fully-recoverable-branch structure is exactly what makes the algebra finitely exact.

Appendix R — Reproducibility

```
.venv/bin/python simulations/classical/m11_lattice_export_density.py # extensivity; a 0.30076
.venv/bin/python simulations/classical/m11_idem_export_density.py # decay bound, exact for
.venv/bin/python simulations/classical/m11_decay_algebra.py # 1D: a=0.3007568, =(3
.venv/bin/python simulations/classical/m11_decay_algebra_general.py # general w; exact reset
.venv/bin/python simulations/classical/m11_decay_algebra_2d.py # 2D strip; a_W converg
.venv/bin/python simulations/classical/m11_continuum_embedding.py # (x)=a((x)), O(1/N),
```

Quantity	Value	Source
AND export $H(X\backslash Y)$	1.188722	Paper A / decay bound
1D density a	0.3007568 ($=(3-\sqrt{5})/2$)	Thm 1
AND density a_w , $w=2,3,4$	0.30076 / 0.56568 / 0.74503	Thm 2
2D a_W , $W=2,3,4$	1.68429 / 1.19329 / 1.03098	§5
continuum total (sample field)	0.29937	§6

Changelog

Date	Entry
2026-07-30	Initial draft: extensivity $\rightarrow$ decay bound $\rightarrow$ 1D theorem $\rightarrow$ general- $w \rightarrow$ 2D strip $\rightarrow$ continuum density $(\mathbf{x})$; classical, no gravity; reproducibility appendix.
